Distributionally Robust LMI Control

Failed Attempts

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Chapter 1

Willems' Fundamental Lemma and Universal Inputs

1.1 Setup and Notation

We consider a discrete–time LTI system

$$x(t+1) = Ax(t) + Bu(t), \qquad y(t) = Cx(t) + Du(t), \tag{1.1}$$

with $u(t) \in \mathbb{R}^m$, $x(t) \in \mathbb{R}^n$ and $y(t) \in \mathbb{R}^p$. The quadruple (A, B, C, D) belongs to

$$\mathcal{M} := \mathbb{R}^{n\times n} \times \mathbb{R}^{n\times m} \times \mathbb{R}^{p\times n} \times \mathbb{R}^{p\times m},$$

and we denote by

$$\mathcal{M}_{\text{cont}} := \{(A, B, C, D) \in \mathcal{M} \mid (A, B) \text{ is controllable}\}$$

the subset of controllable systems.

For an input sequence $u : \mathbb{Z}_{\geq 0} \to \mathbb{R}^m$ and an integer $T \in \mathbb{N}$, we use the stacked vector

$$u_{[0,T-1]} := \begin{bmatrix} u(0)^\top & u(1)^\top & \cdots & u(T-1)^\top \end{bmatrix}^\top.$$

The *Hankel matrix* of depth $k \in \{1, \ldots, T\}$ is

$$H_k\big(u_{[0,T-1]}\big) := \begin{bmatrix} u(0) & u(1) & \cdots & u(T-k) \\ u(1) & u(2) & \cdots & u(T-k+1) \\ \vdots & \vdots & \ddots & \vdots \\ u(k-1) & u(k) & \cdots & u(T-1) \end{bmatrix}. \tag{1.2}$$

[Persistency of Excitation] A finite input sequence $u_{[0,T-1]}$ is said to be *persistently exciting of order* k if the Hankel matrix $H_k(u_{[0,T-1]})$ has full row rank.

We adopt a behavioral viewpoint. The input–output behavior of (1.1) is

$$\mathcal{B}(A, B, C, D) := \Big\{(u, y) : \mathbb{Z}_{\geq 0} \to \mathbb{R}^m \times \mathbb{R}^p \,\Big|\, \exists x(\cdot) \text{ satisfying (1.1)}\Big\}.$$

For a finite horizon length $L \in \mathbb{N}$, the *L–restricted behavior* is the set of all length-L trajectories,

$$\mathcal{B}_L(A, B, C, D) := \left\{ \begin{bmatrix} u_{[0,L-1]} \\ y_{[0,L-1]} \end{bmatrix} \,\middle|\, (u, y) \in \mathcal{B}(A, B, C, D) \right\}.$$

1.2 Willems' Fundamental Lemma

Intuitively, Willems' fundamental lemma states that for a *controllable* LTI system, a single sufficiently long input–output trajectory generated by a *persistently exciting* input completely parametrizes all finite trajectories of the system.

Let $T \in \mathbb{N}$ and $L \in \{1, \dots, T\}$. Suppose we have one experiment on (1.1) producing data

$$\big(u_{[0,T-1]},\, y_{[0,T-1]}\big) \in \mathcal{B}_T(A, B, C, D).$$

Build the Hankel matrices of depth L:

$$H_L\big(u_{[0,T-1]}\big), \qquad H_L\big(y_{[0,T-1]}\big).$$

[Willems' Fundamental Lemma] Let $(A, B, C, D) \in \mathcal{M}_{\text{cont}}$, and let $T, L \in \mathbb{N}$ with $1 \le L \le T$. If $u_{[0,T-1]}$ is persistently exciting of order $n + L$, then

$$\mathcal{B}_L(A, B, C, D) = \operatorname{im} \begin{bmatrix} H_L\big(u_{[0,T-1]}\big) \\ H_L\big(y_{[0,T-1]}\big) \end{bmatrix}. \tag{1.3}$$

In words, any length-L input–output trajectory of the system can be written as a linear combination of columns of the data matrix in (1.3). No explicit model identification is needed once sufficiently rich data are available.

A necessary condition for persistency of excitation of order $n + L$ is the length bound

$$T \;\ge\; (n + L)(m + 1) - 1,$$

which reflects the number of degrees of freedom that must be excited.

1.3 How the Lemma is Used in Data–Driven Control

The lemma can be used as follows:

1. Run *one* sufficiently long experiment on the unknown controllable system with a persistently exciting input u.
2. Construct the Hankel matrices $H_L(u_{[0,T-1]})$ and $H_L(y_{[0,T-1]})$ from the data.
3. Represent any desired length-L trajectory $(\bar{u}_{[0,L-1]}, \bar{y}_{[0,L-1]})$ as

   $$\begin{bmatrix} \bar{u}_{[0,L-1]} \\ \bar{y}_{[0,L-1]} \end{bmatrix} = \begin{bmatrix} H_L(u) \\ H_L(y) \end{bmatrix} g$$

   for some coefficient vector g.
4. Pose control or prediction problems directly in terms of g and the data matrices. Many MPC, simulation, and feedback design schemes are built exactly this way.

This avoids the intermediate step of estimating (A, B, C, D) and instead optimizes directly over behaviors that are consistent with data.

1.4 Universal Inputs and the Main Result of the Paper

The paper introduces the notion of a *universal input*. Fix $T \in \mathbb{N}$ and $L \in \{1, \ldots, T\}$. Consider a finite input sequence $u_{[0,T-1]}$.

[Universal Input for L–Restricted Behavior] The input $u_{[0,T-1]}$ is called *universal for determining the L–restricted behavior* if, for every controllable $(A, B, C, D) \in \mathcal{M}_{\text{cont}}$ and for every output sequence $y_{[0,T-1]}$ generated by (1.1), we have

$$\mathcal{B}_L(A, B, C, D) = \operatorname{im} \begin{bmatrix} H_L\big(u_{[0,T-1]}\big) \\ H_L\big(y_{[0,T-1]}\big) \end{bmatrix}. \tag{1.4}$$

By Theorem 1.2, any input that is persistently exciting of order $n + L$ is universal in the above sense. The key question addressed in the paper is the converse:

*Are there universal inputs that are **not** persistently exciting of order $n + L$?*

The main theorem answers this negatively:

[Universality $\iff$ Persistency of Excitation] Let $T \in \mathbb{N}$ and $L \in \{1, \ldots, T\}$. An input $u_{[0,T-1]}$ is universal for determining the L–restricted behavior if and only if it is persistently exciting of order $n + L$.

The proof is constructive for the "only if" direction: starting from an input that is *not* persistently exciting of order $n + L$, the authors explicitly build a controllable pair (A, B) and an initial state such that the resulting input–state data violate the rank condition required by the lemma. Hence (1.4) fails for that system, so the input is not universal.

In the single–input case ($m = 1$), the paper further shows that if the input is not persistently exciting of order $n + L$, then for *almost all* controllable systems (in a suitable measure–theoretic sense) one can obtain rank–deficient data. Universality is thus extremely fragile without persistency of excitation.

1.5 Implications

The main implications are:

- Persistency of excitation of order $n + L$ is not just a convenient sufficient condition; it is *exactly* what is needed to have a single input that works for *all* controllable systems.
- Online or system–specific experiment design methods can often use shorter experiments, but they lose universality: the resulting input is tailored to one system, not to the entire class of controllable systems.
- For controllable and observable systems with lag ℓ, a universal input of order $n + \ell + 1$ allows, via behavior– based techniques, recovery of a state–space realization (A, B, C, D) up to similarity, by standard subspace identification.

1.6 TikZ Figures

Data–Driven Control Loop Using the Fundamental Lemma

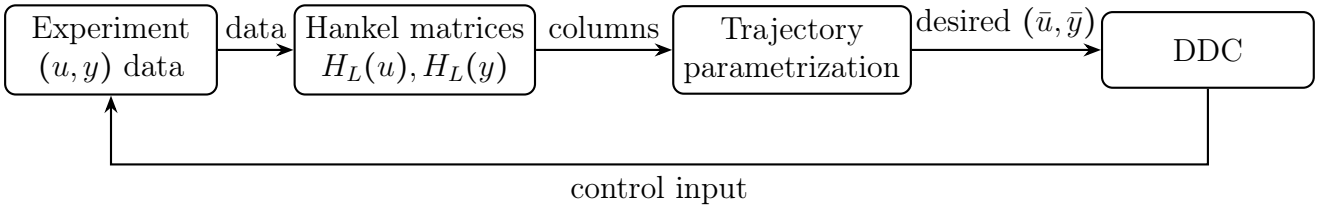

Figure 1.1: Using Willems' lemma: one experiment provides a basis of trajectories for data–driven control design.

Hankel Matrix Structure

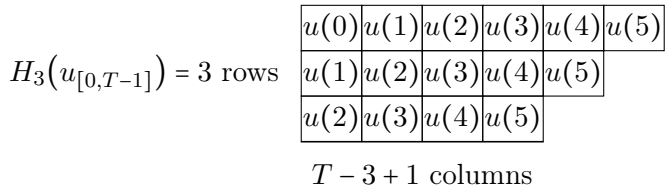

Figure 1.2: Visualization of a Hankel matrix of depth 3 built from a single input sequence.

Converse Construction (Informal)

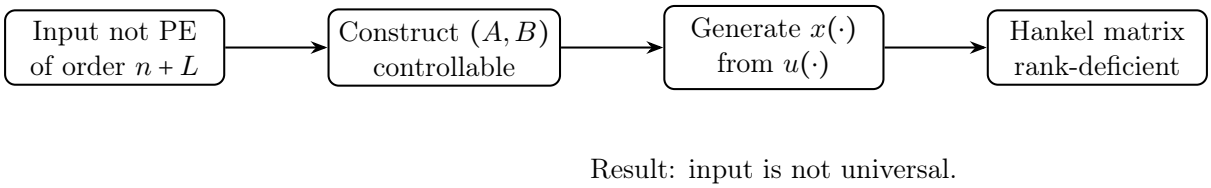

Figure 1.3: Idea of the converse: from a non–PE input one can build a controllable system for which the data violate the rank condition.

1.7 Calculus

We consider an unknown LTI system and aim to design a Direct Data–Driven Controller (DDDC). Let

$$x_e := \begin{bmatrix} x \\ u \end{bmatrix}, \qquad x_{e,k+1} = \underbrace{\begin{bmatrix} A_x & B_u \\ 0 & I \end{bmatrix}}_{A_e} x_{e,k} + \underbrace{\begin{bmatrix} B_w \\ 0 \end{bmatrix}}_{B_e} w_k, \qquad w_k \sim \mathcal{N}(0, \sigma^2).$$

In open loop the system is

$$x_{k+1} = Ax_k + B_u u_k + B_w w_k, \qquad y_k = C_y x_k + D_{yw} w_k.$$

We denote by

$$\gamma = \begin{bmatrix} u \\ y \end{bmatrix}$$

any input–output trajectory. The set of all such trajectories of length L is the behavioral set B_L:

$$B_{L,T} = \left\{ \gamma_{1:T} \in \mathbb{R}^{(n+p)L} \;\middle|\; \exists x_0, \{u_k\} \text{ satisfying the system equations} \right\}.$$

Given measured data $\{u_k, y_k\}_{k=0}^{T-1}$, we form the Hankel matrices

$$H_L(u) = \begin{bmatrix} u_0 & u_1 & \cdots & u_{T-L} \\ u_1 & u_2 & \cdots & u_{T-L+1} \\ \vdots & \vdots & \ddots & \vdots \\ u_{L-1} & u_L & \cdots & u_{T-1} \end{bmatrix}, \qquad H_L(y) \text{ similarly.}$$

Assuming u is persistently exciting (P.E.) of order $n + L$, we have the Willems-parametrization:

$$\forall \gamma = (u, y)\ \exists g : \begin{bmatrix} H_L(u) \\ H_L(y) \end{bmatrix} g = \begin{bmatrix} u \\ y \end{bmatrix}.$$

But the data are noisy, so exact membership fails. We split the Hankel matrices into past/future blocks:

$$\begin{bmatrix} u_p \\ y_p \\ u_f \\ y_f \end{bmatrix} = \begin{bmatrix} H_p(u) \\ H_p(y) \\ H_f(u) \\ H_f(y) \end{bmatrix} g,$$

treating noise via a slack variable δ.

We then consider the optimization problem

$$\min_{g,\delta} \ \|W_p \delta\|_2^2 + \lambda_3 \|g\|_2^2 + \lambda_u \|u_f g - u_f^\star\|_2^2, \qquad W_p \geq 0,\ \lambda_3 > 0,\ \lambda_u \geq 0, \tag{P}$$

subject to

$$\begin{bmatrix} u_p \\ y_p \end{bmatrix} g = \begin{bmatrix} u_p \\ y_p \end{bmatrix} + \delta, \qquad y_f = y_f g, \qquad \|W_p \delta\|_2^2 \leq \eta.$$

However, if the input u is *not* persistently exciting of order $n + L$, then $\begin{bmatrix} H_L(u) \\ H_L(y) \end{bmatrix}$ does not span B_L, and Problem (P) cannot enforce consistency with the unknown system.

1.7.1 Attempt to Replace the Hankel Parametrization with FIR Control

Considering that we can't use this above as substitude of the set $M = (A, B_u, B_w, C_y)$ to then define the augemented LMI system combining $M_c = (A_c, B_c, C_c, D_c)$ into $\mathbf{M} = (\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}, \mathbf{X}, \mathbf{Y}, \mathbf{L}, \mathbf{N}, \mathbf{M}, \mathbf{K})$, we attempt to define a finite impulse-response (FIR) controller:

$$u_k = \sum_{i=0}^{L_c - 1} K_i \, y_{k-i}, \qquad K_i \in \mathbb{R}^{n_u \times n_y}.$$

Stacking over the past/future windows,

$$u_p = T_{y,p\to p}(K)\, y_p + T_{y,p\to f}(K)\, y_f, \qquad u_f = T_{y,f\to p}(K)\, y_p + T_{y,f\to f}(K)\, y_f,$$

where $T_{y,\cdot\to\cdot}(K)$ are truncated Toeplitz matrices.

We then propose the optimization

$$\min_{g,\delta,K} \|W_p z\|_2^2 + \lambda_3 \|g\|_2^2 + \lambda_5 \|W_p \delta\|_2^2 + \lambda_k \|K\|_2^2, \tag{P$_{\text{FIR}}$}$$

subject to

$$\begin{bmatrix} u_p \\ y_p \end{bmatrix} g = \begin{bmatrix} u_p \\ y_p \end{bmatrix} + \delta, \qquad u_f = T_{y,p\to p}(K) y_p + T_{y,f\to f}(K) y_f.$$

1.8 Where the Derivation Deviates from Willems' Fundamental Lemma

The Willems Fundamental Lemma (WFL) states:

> If the input u is persistently exciting of order $n + L$, and the system (A, B) is controllable, then every length-L input–output trajectory of the system belongs to the column span of the Hankel data matrix.

We highlight the steps of the derivation where this requirement is violated.

(1) Assuming Behavioral Parametrization with Noisy Data. The equality

$$\begin{bmatrix} H_L(u) \\ H_L(y) \end{bmatrix} g = \begin{bmatrix} u \\ y \end{bmatrix}$$

is only valid for *noise-free* data and when u is P.E. of order $n + L$.

With noise present, no g exists that satisfies the equation exactly. Replacing the equality by

$$\begin{bmatrix} u_p \\ y_p \end{bmatrix} g = \begin{bmatrix} u_p \\ y_p \end{bmatrix} + \delta$$

is standard in DeePC, but it no longer enforces membership in the behavioral set B_L. It enforces only *approximate* membership modulo the slack δ.

Thus the WFL (Willems' Fundamental Lemma) no longer applies in its exact form.

(2) Loss of Spanning Property When u Is Not P.E. If the input is not P.E. of order $n + L$, the matrix $\begin{bmatrix} H_L(u) \\ H_L(y) \end{bmatrix}$ has insufficient rank. Hence

$$\operatorname{im} \begin{bmatrix} H_L(u) \\ H_L(y) \end{bmatrix} \subsetneq B_L,$$

and the parametrization $(u, y) = \begin{bmatrix} H_L(u) \\ H_L(y) \end{bmatrix} g$ cannot represent all valid system trajectories.

Problem (P) therefore optimizes over trajectories that do *not* correspond to the unknown system.

(3) Replacing the Behavioral Parametrization by FIR Control. The FIR relation

$$u_k = \sum_{i=0}^{L_c-1} K_i y_{k-i}$$

introduces a convolutional structure unrelated to the system behavior captured by the Hankel matrix. The resulting feasible set consists of all pairs (u, y) generated by the chosen FIR controller, but does not ensure that

$$(u, y) \in B_L.$$

Hence Problem ($\mathrm{P}_{\mathrm{FIR}}$) is no longer a data-driven behavioral optimization in the sense of WFL, but a regularized identification problem with a parametric controller.

(4) Toeplitz Constraints Do Not Guarantee Consistency. The constraints of the form

$$u_f = T_{y,p\to p}(K) y_p + T_{y,f\to f}(K) y_f$$

only enforce consistency with the imposed FIR controller, not with the unknown system that generated the data.

Thus WFL cannot be invoked, because the fundamental assumption

"trajectories are consistent with the unknown system"

is not enforced.

Summary.

- The behavioral parametrization via Hankel matrices is valid only for *noise-free* data and P.E. inputs of order $n + L$.
- With noise, it must be relaxed, but the relaxed constraints no longer guarantee consistency with the system behavior.
- Replacing the Hankel parametrization by an FIR controller removes the behavioral foundation entirely; it yields a different optimization problem that does not preserve the guarantees of the Willems Fundamental Lemma.

Chapter 2

On the Failure of Regularized Regression to Preserve LMI Convexity

2.1 Motivation

In data–driven LMI synthesis we often attempt to replace the unknown matrices $(A, B_u, B_w, C_y, D_{yw}, C_z, D_{zu}, D_{zw})$ by estimates obtained from trajectory data. A common approach is to compute regularized least–squares regressions of the form

$$A := X_+ X^\top (XX^\top + \rho_x I)^{-1}, \tag{2.1}$$
$$B_u := X_+ U^\top (UU^\top + \rho_u I)^{-1}, \tag{2.2}$$
$$B_w := X_+ W^\top (WW^\top + \rho_w I)^{-1}, \tag{2.3}$$

and similarly for the output and disturbance matrices,

$$C_y = YX^\top (XX^\top + \rho_x I)^{-1}, \tag{2.4}$$
$$D_{yw} = YW^\top (WW^\top + \rho_w I)^{-1}, \tag{2.5}$$
$$C_z = ZX^\top (XX^\top + \rho_x I)^{-1}, \tag{2.6}$$
$$D_{zu} = ZU^\top (UU^\top + \rho_u I)^{-1}, \tag{2.7}$$
$$D_{zw} = ZW^\top (WW^\top + \rho_w I)^{-1}. \tag{2.8}$$

These expressions arise naturally by solving the Tikhonov regularized identification problems

$$\min_{A, B_u, B_w} \|X_+ - AX - B_u U - B_w W\|_F^2 + \rho_x \|A\|_F^2 + \rho_u \|B_u\|_F^2 + \rho_w \|B_w\|_F^2,$$

whose first–order optimality conditions yield the closed–form pseudoinverse expressions above.

2.2 The Problem: These Inverses Break Convexity

The resulting matrices $A, B_u, B_w, \ldots$ appear inside the closed–loop LMI through the lifted system

$$\mathcal{A} = \begin{bmatrix} AY + B_u N C_y & A + B_u N C_y \\ K & XA + LC_y \end{bmatrix}, \qquad \mathcal{B} = \begin{bmatrix} B_w + B_u N D_{yw} \\ XB_w + LD_{yw} \end{bmatrix},$$

$$\mathcal{C} = \begin{bmatrix} C_z Y + D_{zu} N C_y & C_z + D_{zu} N C_y \end{bmatrix}, \qquad \mathcal{D} = D_{zw} + D_{zu} N D_{yw}.$$

All these blocks contain *rational* functions of the data through terms of the form

$$(XX^\top + \rho_x I)^{-1}, \qquad (UU^\top + \rho_u I)^{-1}, \qquad (WW^\top + \rho_w I)^{-1}.$$

To understand the origin of this LMI, recall that the synthesis problem aims to guarantee closed–loop stability and a disturbance–to–performance bound of the form

$$\|z_k\|_2^2 \le \gamma \|w_k\|_2^2,$$

under the dynamics

$$x_{k+1} = \mathcal{A} x_k + \mathcal{B} w_k, \qquad z_k = \mathcal{C} x_k + \mathcal{D} w_k.$$

Consider the quadratic Lyapunov function $V(x) = x^\top P x$, $P \succ 0$. The standard dissipativity inequality for the bounded–real property is

$$V(x_{k+1}) - V(x_k) + z_k^\top z_k - \gamma w_k^\top w_k < 0.$$

Substituting the system equations yields the quadratic form

$$\begin{bmatrix} x_k \\ w_k \end{bmatrix}^\top \begin{bmatrix} \mathcal{A}^\top P \mathcal{A} - P + \mathcal{C}^\top \mathcal{C} & \mathcal{A}^\top P \mathcal{B} + \mathcal{C}^\top \mathcal{D} \\ \mathcal{B}^\top P \mathcal{A} + \mathcal{D}^\top \mathcal{C} & \mathcal{B}^\top P \mathcal{B} + \mathcal{D}^\top \mathcal{D} - \gamma I \end{bmatrix} \begin{bmatrix} x_k \\ w_k \end{bmatrix} < 0.$$

Applying the Schur complement with respect to the lower–right block gives the equivalent LMI

$$\begin{bmatrix} P\mathcal{A} + \mathcal{A}^\top P & P\mathcal{B} \\ \mathcal{B}^\top P & -\gamma I \end{bmatrix} \prec 0,$$

which is the form commonly used in controller synthesis.

Thus, the block LMI above is simply the discrete–time bounded–real lemma written in Lyapunov form, and it is the natural convex constraint used in all H_∞, dissipativity, and distributionally–robust LMI controllers. Any non–affine dependence of $(\mathcal{A}, \mathcal{B})$ on the decision variables directly destroys the convexity of this constraint.

When these inverses appear multiplied with decision variables X, Y, K, L, N inside an LMI constraint

$$\begin{bmatrix} P\mathcal{A} + \mathcal{A}^\top P & P\mathcal{B} \\ \mathcal{B}^\top P & -\gamma I \end{bmatrix} \prec 0,$$

the dependence of the LMI on the decision variables becomes *nonlinear*. Even worse, the data–dependent inverses introduce nonconvex bilinearities of the form

$$X(XX^\top + \rho_x I)^{-1}, \qquad L(UU^\top + \rho_u I)^{-1}, \qquad N(WW^\top + \rho_w I)^{-1},$$

which immediately destroy the convexity of the synthesis problem. Therefore, Tikhonov–regularized identification cannot be directly plugged into an LMI controller design.

2.3 Attempted Fix: Replacing the Inverse by an Auxiliary Variable

One might attempt to recover convexity by replacing the inverse

$$(XX^\top + \rho_x I)^{-1}$$

with a new positive definite matrix variable $G_x \succ 0$. A tempting definition is

$$G_x := \arg\min_{G \succ 0} \operatorname{tr}\big((XX^\top + \rho_x I)G\big) - \log\det(G),$$

which is known to achieve

$$G_x = (XX^\top + \rho_x I)^{-1}.$$

The idea is then to keep G_x as a variable in the LMI and impose the constraint

$$G_x \succeq (XX^\top + \rho_x I)^{-1} \tag{2.9}$$

or its epigraph approximation.

2.3.1 Why This Fails

The difficulty is structural:

- The inverse map $D \mapsto D^{-1}$ is *matrix concave* on S^n_{++}. Therefore the inequality (2.9) is *nonconvex*.
- To preserve convexity, we would need the hypograph

  $$G_x \preceq D^{-1},$$

  but the hypograph of the inverse is also nonconvex.
- Any linearization or relaxation of $G_x = D^{-1}$ reintroduces nonlinear bilinear terms of the form YG_x, KG_x, LG_x inside the closed–loop matrices.

Thus replacing the inverse by a new variable does *not* fix the problem: the LMI still contains bilinear expressions such as

$$YA = YX_+X^\top G_x, \qquad LB_u = LX_+U^\top G_u,$$

which are nonconvex in the decision variables (Y, L, G_x, G_u).

2.4 Consequences

Even if the auxiliary matrices G_x, G_u, G_w are introduced, the closed–loop matrices $(\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D})$ remain *non-affine* in the decision variables. Hence the data-driven controller synthesis problem takes the nonconvex form:

$$\begin{aligned}
&\text{find } P \succ 0, X, Y, K, L, N, G_x, G_u, G_w \\
&\text{such that} \\
&\qquad \begin{bmatrix} P\mathcal{A}(X, Y, K, L, N, G_x, G_u, G_w) + \mathcal{A}^\top P & P\mathcal{B}(\cdot) \\ \mathcal{B}^\top(\cdot)P & -\gamma I \end{bmatrix} \prec 0.
\end{aligned}$$

This matrix inequality is not convex because the entries of $\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}$ contain nonconvex products of the form

$$XG_x, \qquad YG_x, \qquad LG_u, \qquad NG_w.$$

2.5 Conclusion

Regularized least–squares regression produces closed–form expressions for $(A, B_u, B_w, C_y, D_{yw}, C_z, D_{zu}, D_{zw})$ that depend rationally on the data through inverse Gram matrices. These inverses, when embedded into the closed–loop lifted variables of an LMI-based synthesis method, destroy convexity.

Introducing auxiliary inverse variables via the convex surrogate

$$\min_{G} \ \operatorname{tr}(DG) - \log\det(G)$$

does not resolve the difficulty. The resulting LMIs remain bilinear in the decision variables and therefore nonconvex.

The conclusion is that *regularized regression is incompatible with a convex LMI synthesis pipeline.* One must either work with alternative linear parametrizations or employ explicitly nonconvex optimization methods if identification and synthesis are to be combined.

Chapter 3

Why Existing WFL–Based and SLS–Based Methods Do Not Extend to the Proposed Robust LMI Controller

3.1 Overview

This chapter reviews several influential works on data–driven control and identifies why they cannot be directly employed in the robust dynamic output–feedback problem considered in this thesis. The papers analyzed are:

- Markovsky et al. on the behavioral approach and direct data–driven control [? ?],
- Guo et al. on ambiguity sets using Bregman and Wasserstein distances [?],
- Lian and Jones on the relation between SLS and Willems' Fundamental Lemma [?].

All these methods rely in one way or another on either *state–feedback* or on the existence of a well–defined state trajectory for both plant and controller. The present work, however, requires a robust *dynamic output–feedback* controller whose internal state and closed–loop matrices are *decision variables*. Therefore, the controller state cannot be recovered from data, and open–loop trajectories of the augmented system cannot be constructed, which makes direct application of these methods impossible.

3.2 Behavioral Approach and Willems' Fundamental Lemma

3.2.1 Key result

Willems' Fundamental Lemma (WFL) states that if the measured input is persistently exciting of order $L+n$, then the Hankel matrix $H_L(u,y)$ spans the entire set of admissible length–L trajectories of the unknown LTI system:

$$\operatorname{im} H_L(u,y) = \mathcal{B}_L(A,B,C,D),$$

i.e. one can represent any feasible trajectory using past measured data.

3.2.2 Advantages

- Trajectories are represented without learning (A, B, C, D).
- Convex trajectory optimization is possible for finite–horizon control.
- Exact results hold in the noise–free case.

3.2.3 Limitations for robust LMI synthesis

- The lemma describes *trajectories*, not state–space operators.
- With noise, trajectory constraints become only approximate, not exact.
- The lemma applies to (u, y) but not to internal controller states.
- Dynamic output–feedback requires modeling the internal controller dynamics

  $$x_{c,k+1} = A_c x_{c,k} + B_c y_k, \qquad u_k = C_c x_{c,k} + D_c y_k,$$

  where (A_c, B_c, C_c, D_c) are *unknown* decision variables.
- Hence open–loop trajectories of x_c cannot be generated from data.

Several papers also treat the disturbance as a *measured* signal or as part of the input trajectory, which is incompatible with the setting considered in this thesis where w_k is unmeasured and enters only through (B_w, D_{yw}, D_{zw}). Thus the behavioral approach does not produce the objects needed for robust LMI design.

3.3 Markovsky et al. (2006, 2023)

3.3.1 Contributions

The behavioral framework provides exact and approximate interpolation of trajectories using Hankel matrices and convex surrogates (e.g. nuclear norm). These papers give a comprehensive theoretical foundation for trajectory–based modeling.

3.3.2 Advantages

- Abstract and powerful trajectory characterizations.
- Provides approximate trajectory reconstruction in noisy settings.
- Avoids explicit identification.

3.3.3 Limitations for this thesis

- These methods do not produce state–space matrices (A, B, C, D).
- Approximate fits require nonconvex optimization.
- No form of robust state–space uncertainty suitable for LMIs is produced.
- The behavioral constraints do not handle unmeasured disturbances.

3.4 Guo et al. (2017): Bregman and Wasserstein Ambiguity Sets

3.4.1 Contributions

This work constructs data–driven ambiguity sets using Bregman and Wasserstein distances, producing statistically meaningful distributional uncertainty sets around the empirical disturbance distribution.

3.4.2 Advantages

- Provides principled measures of distributional uncertainty.
- Connects regularization with statistical confidence.
- Characterizes ambiguity sets purely from data.

3.4.3 Limitations

- Ambiguity sets describe distributions, not system matrices.
- The approach does not incorporate dynamic output–feedback.
- Some formulations treat the disturbance as *known* or *measurable*, incompatible with the unmeasured w_k of this thesis.
- No convex LMI formulation is available for dynamic controllers with unknown states.

3.5 Lian & Jones (2021): From SLS to Robust DeePC

3.5.1 Contributions

This paper proves the equivalence between System–Level Synthesis (SLS) and WFL for uncertain LTI systems under full state feedback. SLS expresses closed–loop system responses as

$$(I - ZA)\Phi_x - ZB\Phi_u = I,$$

and derives a controller

$$u_k = \Phi_u \Phi_x^{-1} x_k.$$

3.5.2 Advantages

- Clear link between SLS and the behavioral approach.
- Robust data–driven constraints can be formulated.
- Provides a convex parameterization for state–feedback.

3.5.3 Fundamental incompatibility with this thesis

- SLS requires full state–feedback: $u_k = Kx_k$.
- The controller in this thesis is *dynamic output–feedback*.
- The internal controller state x_c depends on unknown matrices; its open–loop trajectory cannot be computed.
- Even if the plant and controller are augmented, the augmented state–space matrices contain decision variables inside the LMI.
- SLS parametrizations require known system matrices, whereas here they are optimization variables subject to robust uncertainty.

Several examples in the SLS literature also assume that disturbances are either measured or can be injected, which does not align with the unknown, unmeasured disturbance channels in the robust LTI problem here.

3.6 Conclusion

Existing approaches based on WFL, behavioral modeling, Wasserstein ambiguity sets, and SLS do not extend to the robust dynamic output–feedback problem considered in this thesis. The main obstacles are:

1. They require state–feedback or known system matrices.
2. They assume availability of open–loop trajectories of all states, which is impossible when controller internal states depend on unknown matrices.
3. They treat disturbances differently (sometimes as known inputs).
4. They do not provide matrix uncertainty structures compatible with convex LMI robust synthesis.

Therefore these methods cannot be employed as building blocks for the proposed distributionally robust output–feedback LMI controller, and a different framework is required.

Chapter 4

Further Data–Driven and Distributionally Robust Methods

In this chapter we briefly review several data–driven and distributionally robust methods that are closely related to the present work, yet cannot be applied directly to our specific output–feedback, disturbance–aware LMI formulation. For each reference we highlight the core idea, the advantages that are conceptually useful for our framework, and the main mismatches with respect to the problem studied in this thesis.

4.1 Formulas for Data–Driven Control

The work [1] revisits Willems' fundamental lemma and derives a purely data–driven parameterisation of closed–loop system matrices for linear systems. The key result is that, under persistency of excitation, one can express the closed–loop transition matrix in terms of measured state–input trajectories and an auxiliary gain matrix. This parameterisation is then embedded into Lyapunov inequalities to obtain LMI conditions for: (i) state–feedback stabilisation, (ii) output–feedback stabilisation, (iii) LQR design using only data and no explicit model identification.

Pros for our purposes

- Provides an explicit, data–driven closed–loop parameterisation that interfaces well with Lyapunov LMIs.
- Shows clearly how Willems' lemma can be combined with classical LMI design to obtain convex synthesis conditions.
- Includes robustness analysis with respect to noisy data, via additional LMIs that enforce stability for all models consistent with the data.

Limitations for our specific problem

- The main development is for *state–feedback* controllers; the output–feedback case is treated, but in a more restrictive way and without the kind of dynamic, internally stabilising controller structure required in this thesis.

- Noise is modelled through bounded perturbations and quadratic constraints on the data, not via a Wasserstein ambiguity set on the disturbance distribution. Consequently, the robustness guarantees are not directly compatible with a distributionally robust LMI of the type used here.

- The LMIs involve decision variables whose dimensions scale with the state dimension, but do not explicitly separate disturbance channels as in the (A, B_u, B_w) setting adopted in this work. Direct reuse of the formulas would therefore require a substantial reparametrisation of the closed–loop dynamics.

4.2 From Noisy Data to Feedback Controllers via a Matrix S–Lemma

The paper [2] develops a new matrix S–lemma and uses it to design state–feedback controllers from noisy input–state data. The unknown plant is described by $x_{k+1} = Ax_k + Bu_k + w_k$, with bounded process noise. All system matrices consistent with the data and the noise bound are characterised by a quadratic matrix inequality (QMI). The matrix S–lemma is then used to decide whether a candidate state–feedback gain K stabilises *all* such models, leading to nonconservative LMI conditions for quadratic stabilisation, $\mathcal{H}_2$ and $\mathcal{H}_\infty$ control.

Pros for our purposes

- Provides a sharp, nonconservative way to encode noisy data through a QMI and a matrix S–lemma, which is conceptually close to robust LMIs with data–dependent uncertainty sets.

- The LMIs remain of fixed dimension, independent of the length of the data trajectory, which is attractive for large datasets.

- Highlights a clean separation between (i) the set of all models consistent with the data and (ii) the robust controller design on this set.

Limitations for our specific problem

- The controller is *static state–feedback*. Dynamic output–feedback, with an internal controller state that depends on unknown matrices, is not covered.

- Disturbances are treated as bounded additive noise satisfying a QMI. There is no Wasserstein ambiguity set on the disturbance distribution, hence no direct link with the distributionally robust radius γ used in our LMI.

- The open–loop model set is defined in (A, B) space; there is no explicit disturbance channel B_w and no lifted closed–loop structure of the type $(\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D})$ used here.

4.3 Extended State Framework for Direct Data–Driven Control

The thesis [3] proposes an extended–state representation that embeds the plant and a model–based controller into a larger linear system. Using Willems' lemma, this extended system is then represented directly from input–output data, and stabilising controllers are designed by working on a reduced–order extended state. The focus is on noise–free data and on constructing a data–based state–feedback law acting on the extended state.

Pros for our purposes

- Shows how to embed a dynamic output–feedback controller into an extended state and still obtain a data–based representation of the closed–loop dynamics.
- Provides a clear recipe to switch between model–based and data–based extended representations, which is useful conceptually for hybrid designs.
- Demonstrates that dynamic controllers can be handled in a purely data–driven way when the extended state is directly reconstructible from trajectories.

Limitations for our specific problem

- The framework assumes (essentially) noise–free trajectories when deriving the data–based extended state model. Disturbances are not modelled through a Wasserstein ambiguity set nor through explicit stochastic uncertainty.
- Even though the controller is dynamic, its internal state is defined in terms of known controller polynomials. In our setting, the internal controller state depends on unknown matrices to be found via LMIs, and cannot be reconstructed from open–loop data.
- The main design results are not phrased as robust LMIs with explicit disturbance channels; reusing them would require a full reformulation of the disturbance model and of the distributionally robust constraints.

4.4 Data–Driven Distributionally Robust Control of Partially Observable Jump Linear Systems

In [4], the authors consider a partially observable Markov jump linear system with unknown mode transition probabilities. A receding–horizon estimator recovers a subset of mode sequences and the associated continuous states. Based on the resulting dataset, ambiguity sets over the transition probabilities are built as ℓ_1–balls around the empirical distributions. A distributionally robust output–feedback controller is then designed so as to stabilise the MJLS in the mean–square sense for all probability vectors in these polytopic ambiguity sets.

Pros for our purposes

- Provides a full pipeline from data to ambiguity sets to distributionally robust controller, explicitly in an output–feedback setting.
- The ambiguity sets shrink as more data are collected, leading to an adaptive, less conservative controller.
- The use of polytopic ambiguity sets shows how probability–simplex uncertainty can be incorporated into robust control LMIs.

Limitations for our specific problem

- The uncertainty is on the *mode transition probabilities* of an MJLS, not on an additive disturbance process entering through B_w. The ambiguity sets are ℓ_1–balls on the simplex, not Wasserstein balls on continuous disturbances.
- The system is essentially state–feedback per mode, with output measurements used for mode estimation. This structure is different from the fixed LTI output–feedback architecture with internal controller state considered in this thesis.
- The robust LMIs are tailored to finite–dimensional uncertainty in the transition matrix; they do not generalise directly to Wasserstein ambiguity sets over disturbance trajectories.

4.5 Data–Driven DRO Using the Wasserstein Metric

The work [5] develops a general framework for data–driven distributionally robust optimisation where the ambiguity set is a Wasserstein ball around the empirical distribution. For a broad class of loss functions and decision sets, the worst–case expected cost admits a tractable finite–dimensional reformulation. Finite–sample performance guarantees and asymptotic consistency are established under a light–tailed assumption on the unknown distribution.

Pros for our purposes

- Provides the fundamental nonasymptotic guarantees and concentration bounds that justify Wasserstein ambiguity sets and the scaling of the radius γ with the sample size.
- The dual reformulations clarify how Wasserstein DRO induces regularisation terms in the optimisation problem, which is conceptually useful when interpreting data–driven robust LMIs.
- The framework is completely general and can, in principle, be coupled with control design as long as the closed–loop loss fits the required structure.

Limitations for our specific problem

- The paper treats a *static* optimisation problem with respect to a decision variable x, not a dynamic output–feedback controller embedded in LMIs.
- The disturbance enters only through the objective function $h(x, \xi)$; there is no explicit state–space realisation with disturbance channels (B_w, D_{zw}, D_{yw}) as in the present work.
- The tractable reformulations are not directly in LMI form for closed–loop stability constraints. To use them in our synthesis problem would require a nontrivial lifting and coupling with Lyapunov inequalities.

4.6 Cost–Aware Ambiguity Sets

The article [6] introduces Cost–Aware DRO (CADRO), where the ambiguity set on a finite–dimensional probability vector p is shaped using information about the cost function. Instead of a symmetric metric ball, the ambiguity set is elongated along directions in which the expected cost is most sensitive, aiming to reduce conservatism while still guaranteeing high–confidence performance bounds.

Pros for our purposes

- Shows that generic metric balls (including Wasserstein) are not the only meaningful choice for ambiguity sets: cost structure can be exploited to obtain tighter, yet still safe, sets.
- Provides a constructive algorithm to build admissible ambiguity sets that preserve finite–sample guarantees and asymptotic consistency.
- Conceptually suggests that one could design disturbance ambiguity sets tailored to the closed–loop LMI cost used in robust control.

Limitations for our specific problem

- The ambiguity sets live on a *finite* probability simplex and are tuned with respect to a static cost. Our disturbance process is continuous–valued and enters a dynamical state–space system.
- The resulting DRO problems are not expressed as Lyapunov LMIs for closed–loop stability, but as convex programs over x and p. Bridging this to a robust LMI for (A, B_u, B_w) would require a substantial reformulation.
- In several control–oriented examples, disturbances are treated at the level of stochastic cost, not as an explicit additive disturbance with Wasserstein ambiguity compatible with our synthesis conditions.

4.7 Comparison

Tables 4.1 and 4.2 summarise the main features, advantages and limitations of these methods with respect to the robust output–feedback LMI problem considered in this thesis.

Table 4.1: Comparison of related data–driven / DRO methods (Part I)

Reference	Main setting	Noise / ambiguity model	Main pros / main limitations for this thesis
De Persis & Tesi [1]	LTI, state & output feedback, data–driven LMIs	Bounded noise via quadratic constraints on data; no Wasserstein set	Pros: explicit closed–loop parametrisation compatible with LMIs. Limit: mainly state–feedback; robustness not expressed via Wasserstein ambiguity on disturbances.
van Waarde et al. [2]	LTI with state–feedback, noisy state/input data	Bounded additive process noise described by a QMI; matrix S–lemma	Pros: nonconservative robust LMIs; data length independent of decision size. Limit: static state–feedback only; no explicit B_w channel or Wasserstein ambiguity.
Ravikkumar [3]	Extended–state framework, dynamic controller	Essentially noise–free trajectories; robustness not distributional	Pros: shows how to embed dynamic controllers into an extended state and apply Willems' lemma. Limit: internal controller state must be known; cannot handle our unknown dynamic output–feedback structure with Wasserstein disturbance modelling.

Table 4.2: Comparison of related data–driven / DRO methods (Part II)

Reference	Main setting	Noise / ambiguity model	Main pros / main limitations for this thesis
Schuurmans & Patrinos (MJLS) [4]	Partially observable MJLS, output feedback	ℓ_1–balls on transition probabilities (polytopic ambiguity)	Pros: full pipeline from data to ambiguity sets to DR controller in output–feedback form. Limit: uncertainty is on mode probabilities, not on an additive disturbance with Wasserstein radius; structure incompatible with our LTI (A, B_u, B_w) model.
Esfahani & Kuhn [5]	Static optimisation with Wasserstein DRO	Wasserstein ball around empirical distribution, with finite–sample guarantees	Pros: rigorous justification for Wasserstein ambiguity sets and radius scaling. Limit: no explicit state–space or LMI structure; disturbances appear only through a static loss, not through dynamic output–feedback synthesis.
Schuurmans & Patrinos (CADRO) [6]	Generic DRO with cost–aware ambiguity sets	Cost–shaped ambiguity sets on a probability simplex (not necessarily Wasserstein)	Pros: demonstrates how cost structure can shape ambiguity sets and reduce conservatism. Limit: finite–dimensional probability vectors, static decisions; not directly applicable to dynamic output–feedback LMIs with disturbance channels.

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